

UdyamPulse

An explainable MSME Financial Health Card built for IDBI Innovate 2026.

Team name	Looper
Team leader name	Bhunesh Bansal
Problem Statement	Problem Statement 3 — Financial Health Score

Consented alternate data, turned into a decision a bank can defend.

01 What it is.

An explainable MSME Financial Health Card that scores businesses from consented alternate data.

02 Who it's for.

New-to-Credit (NTC) and New-to-Bank (NTB) MSMEs that traditional bureau-based underwriting rejects outright.

03 Inputs.

GST filings, UPI transaction trails, Account Aggregator bank statements, EPFO records.

04 Outputs.

A 0–100 score, an A–E grade, an eligible credit limit, and a plain-language improvement plan.

Explainability and auditability, by default.

DIFFERENTIATION

Beyond a number

Most alternate-data scoring tools return a score. UdyamPulse returns a decision a bank can defend to a regulator and a borrower can act on.

APPROACH

Two engines, one verdict

Independent, cross-checkable scoring methods run side by side and show explicitly what bureau-only underwriting would have decided vs. alternate data.

USP

The regulatory gap

Of 127 AI-using regulated entities surveyed, only 15 use SHAP/LIME explainability and only 18 keep audit logs (RBI). UdyamPulse ships both by default.

What the health card actually does.

5-Pillar Health Score

Liquidity, Discipline, Momentum, Leverage, Digital Footprint → 0–100 score, A–E grade, eligible limit.

Explainable ML + Shapley

A trained model with exact per-feature Shapley attribution behind every decision — no black box.

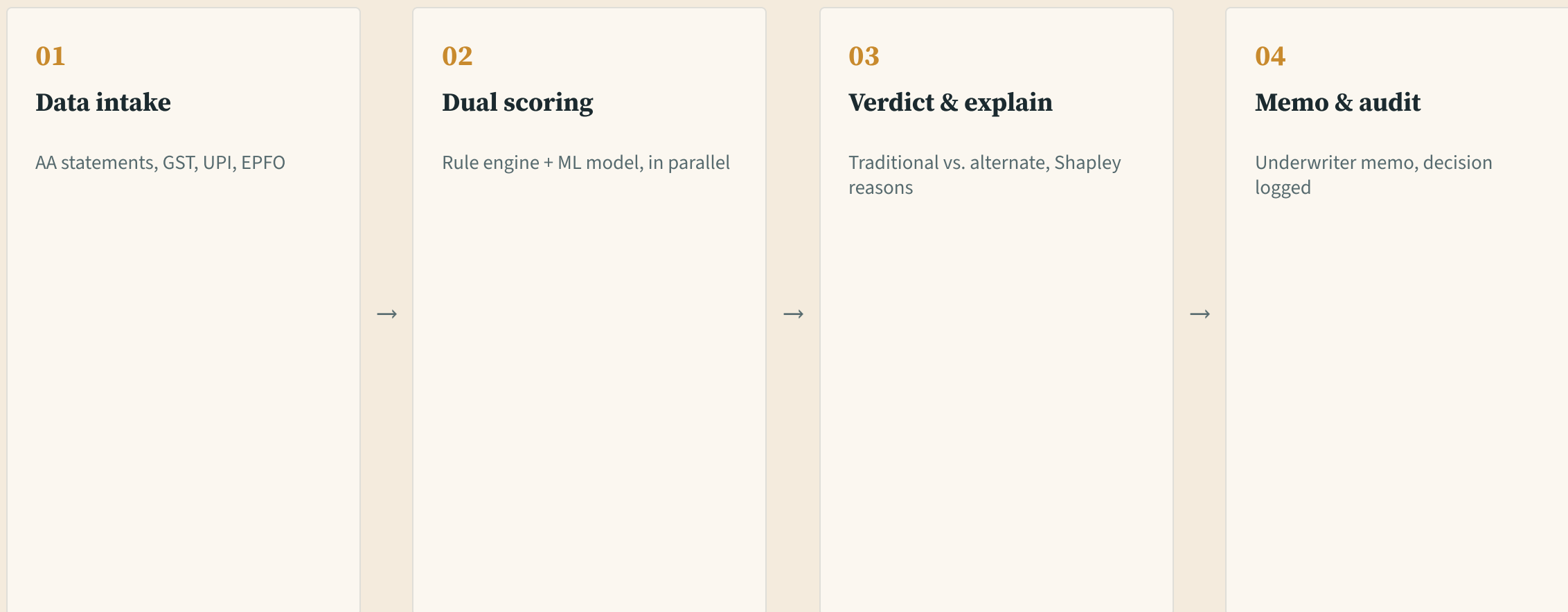
Traditional vs. Alternate Verdict

Side-by-side comparison showing the rejected-to-approved reversal for thin-file businesses.

Improvement Plan + Audit Log

A concrete action plan for borrowers, and an append-only audit trail for every decision made.

From consented data to an auditable decision.



The rejected → approved moment.

The NTC hero (Shree Ganesh Textiles) has zero bureau history — declined instantly by a traditional process, approved on alternate data by UdyamPulse.

TRADITIONAL (BUREAU-ONLY)

Rejected

No credit bureau history on file (New-to-Credit / New-to-Bank).

UDYAMPULSE (ALTERNATE DATA)

Approved

Grade A · Score 86/100 · Eligible limit ₹27,00,000.

One service. Four inspectable control planes.

Operational Edge

Request IDs, body ceilings, readiness probes, security headers, rate controls, and a non-root container.

Decision + Evidence

Five transparent pillars, calibrated PD, exact TreeSHAP attribution, policy checks, and reason codes.

Temporal Validation

Dated outcomes, 365-day maturity, chronological development/calibration/OOT, PSI, KS, AUC, and Gini.

Promotion + Audit

Pseudonymous hash-chain evidence and fail-closed pilot startup until model, OOT, identity, and storage pass.

A reproducible, inspectable open stack.

Backend	Python, FastAPI, Pydantic, pytest
ML layer	Calibrated monotonic XGBoost champion with native exact TreeSHAP; calibrated logistic fallback
Frontend	Accessible, deep-linkable HTML / CSS / JS cockpit with no build step
Deployment & CI	Non-root Docker runtime, readiness probe, Render Blueprint, two-job GitHub Actions
Stage 2 targets	Supply IDBI sandbox records to the shipped outcome/temporal gates; add SSO/KMS and durable shared audit storage

Near-zero to prototype. Scoped jointly for Stage 2.

The prototype runs entirely on an open-source stack and free-tier hosting. Stage 2 production cost scales with transaction volume and is best scoped jointly with IDBI during the sandbox mentorship phase, once real data volume is known.

The live underwriter cockpit.

Captured from the verified submission build: portfolio impact, case queue, explainable health card, audit controls, source map, and borrower plan in one surface.

UdyamPulse

MSME credit review, without hiding the audit trail.

Up

Live API / xgboost pd proxy v1 model

/ Public synthetic cohort with sandbox feed contracts

Decision

Evidence

Governance

Proof

Sources

Borrower file

Shree Ganesh Textiles (New-to-Credit) ▼

Shree Ganesh Textiles (New-to-Credit) Textiles / Surat A grade / ₹27,00,000	Patel Wholesale Traders Wholesale / Ahmedabad B grade / ₹32,80,000	City Corner Retail Retail / Nagpur D grade / ₹1,80,000	Sunrise Auto Parts Auto Components / Pune C grade / ₹7,75,000	Himalayan H Handicrafts / D B grade /
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UP-ntc-hero

Textiles / Surat

New-to-Credit / New-to-Bank

Bureau

rejection

Low risk

A

Live first viewport: 2 NTC rescues, Rs 30,80,000 unlocked, traditional rejected vs. UdyamPulse approved.

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Decision pack: grade A, score 86/100, Shapley attribution, memo, and guardrails.

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Governance rail: controls, validation metrics, pilot KPIs, fairness slices, audit count, and 5 data-source signals.

Verified, not just claimed.

70/70

Automated tests passing

0.750

Untouched holdout AUC

0.499

Holdout Gini

0.012

Calibration error

Measured on a locked 4,500-row public-proxy holdout: AUC 0.7497 (95% bootstrap CI 0.7314-0.7678), KS 0.4225, Brier 0.1415, and PSI 0.0003. This is cross-sectional evidence, not IDBI calibration or true out-of-time proof.

Stage 2 gates are live. Bank data is next.

Phase 1

Sandbox data intake

Connect authenticated feeds, profile distributions, and validate dated outcomes through /sandbox/pilot-readiness.

Phase 2

IDBI-calibrated champion

Retrain on mature chronological cohorts and promote only when the shipped deployment gate confirms true OOT evidence.

Phase 3

Fairness & monitoring

Run the implemented fairness gap, PSI, and reason-code stability checks on production-scale data.

Phase 4

Bedrock-powered memos

Enable the optional AWS Bedrock provider in the pilot environment, with deterministic fallback.

Where to find everything.

→ GitHub Public Repository <https://github.com/bansalbhunesh/id>

→ Recorded Walkthrough github.com/bansalbhunesh/id/blob/main/docs/demo.webm

→ Final Product Link <https://id-ysm9.onrender.com>